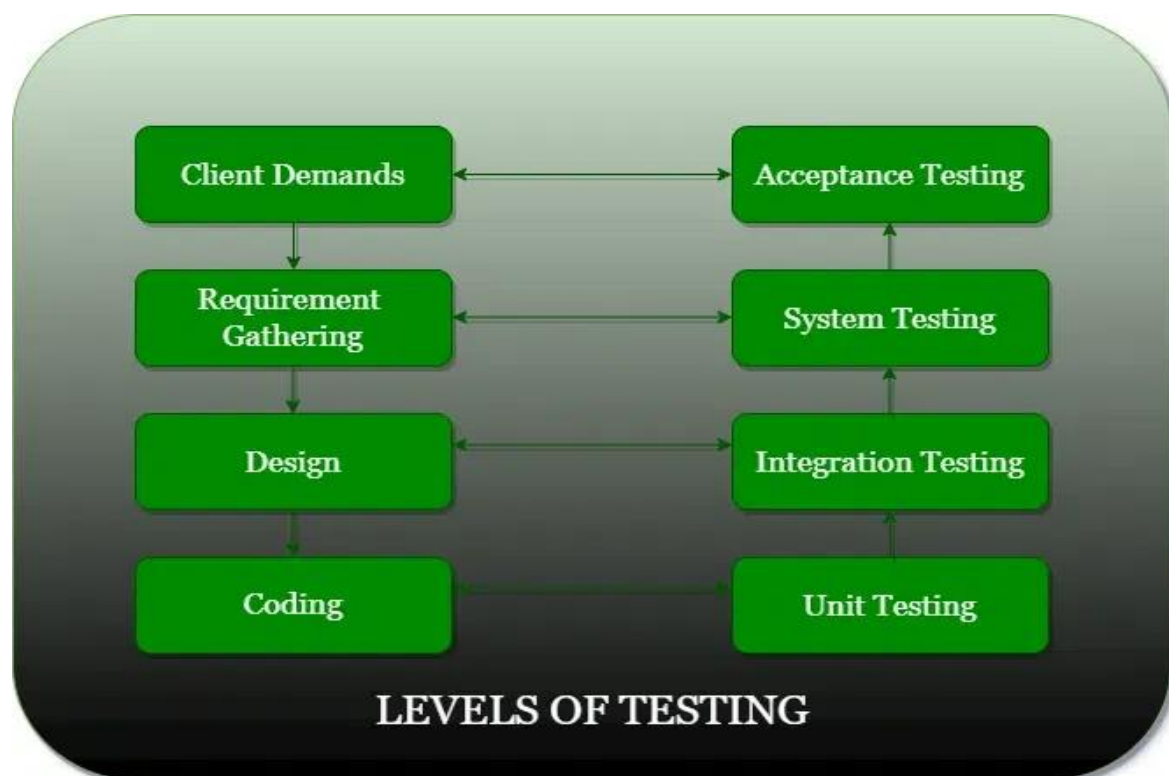


Levels of Software Testing

Software Testing is an activity performed to identify errors so that errors can be removed to obtain a product with greater quality. To assure and maintain the quality of software and to represent the ultimate review of specification, design, and coding, Software testing is required. There are different levels of testing:

1. **Unit Testing**: In this type of testing, errors are detected individually from every component or unit by individually testing the components or units of software to ensure that they are fit for use by the developers. It is the smallest testable part of the software.
2. **Integration Testing**: In this testing, two or more modules which are unit tested are integrated to test i.e., technique interacting components, and are then verified if these integrated modules work as per the expectation or not, and interface errors are also detected.
3. **System Testing**: In system testing, complete and integrated Software's are tested i.e., all the system elements forming the system are tested as a whole to meet the requirements of the system.
4. **Acceptance Testing**: This is a kind of testing conducted to ensure that the requirements of the users are fulfilled before its delivery and that the software works correctly in the user's working environment.

These tests can be conducted at various stages of software development. The levels of testing along with the corresponding software development phase are shown in the following diagram:



Difference between Alpha and Beta Testing

Alpha Testing and Beta Testing are part of software development life cycle. Alpha and Beta testing are types of User Acceptance Testing (UAT). Alpha Testing happens in the early stages of SDLC and is performed internally by the development team while Beta testing happens later in the SDLC, just before product release, and is performed by potential customers, industry experts, or public beta testers.

What is Alpha Testing?

[Alpha Testing](#) is a type of software testing performed to identify bugs before releasing the product to real users or to the public. Alpha Testing is one of the user acceptance tests. It is the first stage of software testing, during which the internal development team tests the program before making it available to clients or people outside the company.

Key points to remember:

Following are the key points related to Alpha Testing:

- **Work Done by Developers:** The internal development team, which consists of developers and testers, usually conducts alpha testing in a controlled setting.
- **Goal:** Finding and fixing bugs, flaws, and usability issues is the main goal before releasing the product for external users or wider testing.
- **Little User Engagement:** Alpha testers are few in number and frequently comprise members of the development team or those intimately connected to the project.
- **Environment:** Alpha testing is typically carried out in a development environment or laboratory that resembles actual settings.

What is Beta Testing?

[Beta Testing](#) is performed by real users of the software application in a real environment. Beta testing is one type of User Acceptance Testing. A pre-release version of the product is made available for testing to a chosen set of external users or customers during the second phase of software testing.

Key Points to remember:

Following are the key points related to Beta Testing:

- **Actions Taken by Users:** Customers or other users outside the development team participate in beta testing. The software is available to these users prior to its official release.
- **Goal:** The primary objective is to get input from actual users in order to find any bugs, usability difficulties, or areas that need to be improved before the product is formally released.
- **Greater User Participation:** In order to capture a variety of viewpoints, a larger number of users—including a varied range, can serve as beta testers.

- **Environment:** Real-world settings are used for beta testing to simulate how users will interact with the program while performing daily duties.

Difference between Alpha and Beta Testing:

The difference between Alpha and Beta Testing is as follows:

Parameters	Alpha Testing	Beta Testing
Technique Used	Alpha testing uses both white box and black box testing.	Beta testing commonly uses black-box testing.
Performed by	Alpha testing is performed by testers who are usually internal employees of the organization.	Beta testing is performed by clients who are not part of the organization.
Performed at	Alpha testing is performed at the developer's site.	Beta testing is performed at the end-user of the product.
Reliability and Security	Reliability and security testing are not checked in alpha testing.	Reliability, security and robustness are checked during beta testing.
Ensures	Alpha testing ensures the quality of the product before forwarding to beta testing.	Beta testing also concentrates on the quality of the product but collects users input on the product and ensures that the product is ready for real time users.
Requirement	Alpha testing requires a testing environment or a lab.	Beta testing doesn't require a testing environment or lab.
Execution	Alpha testing may require a long execution cycle.	Beta testing requires only a few weeks of execution.

Parameters	Alpha Testing	Beta Testing
Issues	Developers can immediately address the critical issues or fixes in alpha testing.	Most of the issues or feedback collected from the beta testing will be implemented in future versions of the product.
Test Cycles	Multiple test cycles are organized in alpha testing.	Only one or two test cycles are there in beta testing.